Event Consistency in Continuous-Time Inventory Routing

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Abstract. Continuous-time inventory routing integrates vehicle movements with inventory trajectories that evolve between visits. Repeated replenishments require all inventory constraints to be evaluated with respect to a single order of delivery events. We show that pairwise precedence variables without transitivity can violate this requirement and admit spurious feasible solutions. Using the event-copy model of Wang et al. (2025) as the reference, we construct a one-customer instance in which every delivery must occur at time 5. The terminal inventory requirement calls for 20 units of replenishment, whereas the tank can receive at most 15 units at that time. A pairwise-precedence formulation admits four deliveries of five units each, with total travel cost 8, even though the resulting inventory reaches 105 in a tank with capacity 100. The instance isolates event consistency as a necessary condition for valid continuous-time inventory-routing formulations.

Keywords. continuous-time inventory routing; repeated delivery; event sequencing; tank replenishment; mixed-integer linear programming

1. Problem Motivation

Business-level description. *A supplier distributes industrial gas, fuel, or liquid raw materials to customers that consume the product continuously. Each customer stores inventory in a tank subject to lower and upper operating limits and a terminal reserve requirement. Vehicles may serve multiple customers, return to the depot to reload, and visit the same customer more than once. The supplier determines vehicle routes, visit times, and delivery quantities to minimize travel cost while maintaining feasible inventory levels throughout the planning horizon.*

Initial and terminal inventory balances alone do not establish continuous-time feasibility because inventory may violate an operating limit between these two observations. A valid formulation must therefore explicitly represent delivery times, quantities, and the resulting inventory trajectory. The need for such an explicit representation is central to the continuous-time inventory-routing problem studied by Lagos et al. (2020), which incorporates continuous consumption, capacitated vehicles, repeated customer visits, and multiple depot trips. Wang et al. (2025) further introduce terminal inventory targets and develop a compact mixed-integer linear programming formulation based on ordered visit copies, which we adopt as the reference formulation.

The ordered-copy structure becomes essential when a customer receives multiple deliveries. The inventory level at each visit depends on the deliveries completed earlier, so the routing, timing, and inventory constraints must share a common event sequence. Pairwise precedence relations that do not collectively define such a sequence can cause different inventory constraints to rely on incompatible delivery histories, thereby admitting a solution that does not correspond to any feasible physical inventory trajectory.

2. Problem Definition

Let $G = (V, E)$ be a complete directed network, where $V = \{0\} \cup C$ is the node set, node 0 represents the depot, and C is the set of customers. The arc set is $E = \{(i, j) \mid i, j \in V,\ i \neq j\}$. Let $[0, H]$ be the planning horizon, where $H > 0$. Each customer $i \in C$ has initial inventory I_i^0, lower and upper inventory limits L_i and U_i, a constant consumption rate $\rho_i > 0$, a terminal inventory target F_i, and a maximum number of visits $n_i \geq 1$. We assume $L_i \leq I_i^0 \leq U_i$, $L_i \leq F_i \leq U_i$, and $F_i + \rho_i H - I_i^0 > 0$.

The depot has unlimited inventory and initially holds $m \geq 1$ identical vehicles, each with capacity $Q > 0$. A vehicle may complete multiple depot-to-depot trips but must return to the depot by time H. Each arc $(i, j) \in E$ has travel cost $c_{ij} \geq 0$ and travel time $\tau_{ij} > 0$. The cost and travel-time matrices satisfy the directed triangle inequality. Loading at the depot is instantaneous, and waiting incurs no cost.

A vehicle may remain at a customer while dispensing product. Positive-duration visits by different vehicles may not overlap at the same customer. Zero-duration visits may occur at the same clock time, provided that they admit a consistent event order.

For each customer $i \in C$, let $I_i(t)$ denote its inventory at time $t \in [0, H]$. A feasible replenishment plan satisfies $L_i \leq I_i(t) \leq U_i$ for all $i \in C$ and $t \in [0, H]$, and $I_i(H) \geq F_i$ for all $i \in C$. The objective is to minimize total vehicle travel cost.

3. Reference Formulation

We adopt the compact event-copy formulation in Section 3 of Wang et al. (2025). For each customer $i \in C$, let $R_i = \{1, \ldots, n_i\}$ be the set of ordered visit copies. Event node (i, α) represents the αth potential visit to customer i, where $\alpha \in R_i$. The pair $(0, 1)$ denotes the unique depot event node. The set of event nodes is $\mathcal{N} = \{(0, 1)\} \cup \{(i, \alpha) \mid i \in C,\ \alpha \in R_i\}$.

Let $D = \{0, 1\}$ denote the set of travel modes. Mode 0 represents direct travel between two physical locations. Mode 1 represents travel from a customer to the depot for reloading and then to the next customer. The corresponding event-arc sets are

$$\mathcal{E}^0 = \{(i, \alpha, j, \beta) \mid (i, \alpha), (j, \beta) \in \mathcal{N},\ i \neq j\},$$
$$\mathcal{E}^1 = \{(i, \alpha, j, \beta) \mid i, j \in C,\ \alpha \in R_i,\ \beta \in R_j,\ i \neq j \text{ or } \alpha < \beta\}.$$

For mode $d \in D$, let c_{ij}^d and τ_{ij}^d denote the travel cost and time from i to j. Direct travel uses $c_{ij}^0 = c_{ij}$ and $\tau_{ij}^0 = \tau_{ij}$, while depot-reload travel uses $c_{ij}^1 = c_{i0} + c_{0j}$ and $\tau_{ij}^1 = \tau_{i0} + \tau_{0j}$.

For event node $(i,\alpha) \in \mathcal{N}$ and mode $d \in D$, define its predecessor and successor sets by

$$\delta_d^-(i,\alpha) = \{(j,\beta) \in \mathcal{N} \mid (j,\beta,i,\alpha) \in \mathcal{E}^d\},$$
$$\delta_d^+(i,\alpha) = \{(j,\beta) \in \mathcal{N} \mid (i,\alpha,j,\beta) \in \mathcal{E}^d\}.$$

For every event arc $(i,\alpha,j,\beta) \in \mathcal{E}^d$, binary variable $x^d_{i\alpha j\beta}$ equals one if that arc is selected in mode d and zero otherwise. For every customer $i \in C$ and copy $\alpha \in R_i$, binary variable $y_{i\alpha}$ equals one if visit copy (i,α) is active.

Nonnegative variables $a_{i\alpha}$ and $b_{i\alpha}$ denote the arrival and departure times of visit copy (i,α), and $q_{i\alpha}$ denotes the quantity delivered during that visit. For each direct event arc $(i,\alpha,j,\beta) \in \mathcal{E}^0$, variable $f_{i\alpha j\beta} \geq 0$ denotes the vehicle load carried on the arc. Variable $\ell_{i\alpha} \geq 0$ denotes the quantity loaded at the depot before a mode-1 arrival at event node (i,α).

For each ordered pair of event nodes connected by at least one travel mode, variable $\widetilde{a}_{i\alpha j\beta} \geq 0$ represents the arrival time at successor event node (j,β) when the corresponding connection is selected and equals zero otherwise. Let $\underline{a}_{i\alpha}$ and $\overline{a}_{i\alpha}$ denote valid lower and upper bounds on the arrival time of visit copy (i,α). We use $\underline{a}_{i\alpha} = \tau_{0i}$ and $\overline{a}_{i\alpha} = H - \tau_{i0}$.

The formulation is

$$\min \quad \sum_{d\in D} \sum_{(i,\alpha,j,\beta)\in\mathcal{E}^d} c^d_{ij} x^d_{i\alpha j\beta}, \tag{1}$$

$$\text{subject to} \quad \sum_{d\in D} \sum_{(j,\beta)\in\delta_d^-(i,\alpha)} x^d_{j\beta i\alpha} = y_{i\alpha} \qquad \forall i \in C,\ \alpha \in R_i, \tag{2}$$

$$\sum_{d\in D} \sum_{(j,\beta)\in\delta_d^+(i,\alpha)} x^d_{i\alpha j\beta} = y_{i\alpha} \qquad \forall i \in C,\ \alpha \in R_i, \tag{3}$$

$$\sum_{(j,\beta)\in\delta_0^+(0,1)} x^0_{0,1,j,\beta} \leq m, \tag{4}$$

$$\sum_{(i,\alpha)\in\bigcup_{d\in D}\delta_d^-(j,\beta)} \widetilde{a}_{i\alpha j\beta} = a_{j\beta} \qquad \forall j \in C,\ \beta \in R_j, \tag{5}$$

$$\underline{a}_{i\alpha} y_{i\alpha} \leq a_{i\alpha} \leq \overline{a}_{i\alpha} y_{i\alpha} \qquad \forall i \in C,\ \alpha \in R_i, \tag{6}$$

$$a_{i\alpha} \leq b_{i\alpha} \qquad \forall i \in C,\ \alpha \in R_i, \tag{7}$$

$$b_{i\alpha} + \sum_{d\in D} \sum_{(j,\beta)\in\delta_d^+(i,\alpha)} \tau^d_{ij} x^d_{i\alpha j\beta} \leq \sum_{(j,\beta)\in\bigcup_{d\in D}\delta_d^+(i,\alpha)} \widetilde{a}_{i\alpha j\beta} \qquad \forall i \in C,\ \alpha \in R_i, \tag{8}$$

$$y_{i,\alpha+1} \leq y_{i\alpha} \qquad \forall i \in C,\ \alpha = 1,\ldots,n_i - 1, \tag{9}$$

$$b_{i\alpha} \leq a_{i,\alpha+1} + (H - \tau_{i0})(y_{i\alpha} - y_{i,\alpha+1}) \qquad \forall i \in C,\ \alpha = 1,\ldots,n_i - 1. \tag{10}$$

Constraints (2) and (3) link visit activation to the selected incoming and outgoing event arcs. Constraint (4) limits the number of vehicles departing from the depot. Constraints (5)–(8) propagate arrival times through the selected vehicle itineraries. Constraints (9) and (10) place active copies first and impose their chronological order.

The auxiliary arrival variables satisfy

$$\underline{a}_{j\beta} x^0_{0,1,j,\beta} \leq \widetilde{a}_{0,1,j,\beta} \leq \overline{a}_{j\beta} x^0_{0,1,j,\beta} \qquad \forall (0,1,j,\beta) \in \mathcal{E}^0, \tag{11}$$

$$0 \leq \widetilde{a}_{i\alpha j\beta} \leq \overline{a}_{j\beta} \sum_{\substack{d\in D \\ (i,\alpha,j,\beta)\in\mathcal{E}^d}} x^d_{i\alpha j\beta} \qquad \forall (i,\alpha,j,\beta) \in \mathcal{E}^0 \cup \mathcal{E}^1,\ i,j \in C, \tag{12}$$

$$0 \leq \widetilde{a}_{i\alpha 0,1} \leq H x^0_{i\alpha 0,1} \qquad \forall (i,\alpha,0,1) \in \mathcal{E}^0. \tag{13}$$

These constraints set each auxiliary arrival variable to zero when the associated event connection is not selected. Together with constraints (5) and (8), they propagate time along each selected route.

The product-flow and vehicle-capacity constraints are

$$\sum_{(i,\alpha)\in\delta_0^-(j,\beta)} f_{i\alpha j\beta} + \ell_{j\beta} = q_{j\beta} + \sum_{(i,\alpha)\in\delta_0^+(j,\beta)} f_{j\beta i\alpha} \qquad \forall j \in C,\ \beta \in R_j, \tag{14}$$

$$f_{i\alpha 0,1} = 0 \qquad \forall (i,\alpha,0,1) \in \mathcal{E}^0, \tag{15}$$

$$0 \leq f_{i\alpha j\beta} \leq Q x^0_{i\alpha j\beta} \qquad \forall (i,\alpha,j,\beta) \in \mathcal{E}^0, \tag{16}$$

$$0 \leq \ell_{j\beta} \leq Q \sum_{(i,\alpha)\in\delta_1^-(j,\beta)} x^1_{i\alpha j\beta} \qquad \forall j \in C,\ \beta \in R_j, \tag{17}$$

$$0 \leq q_{i\alpha} \leq Q y_{i\alpha} \qquad \forall i \in C,\ \alpha \in R_i. \tag{18}$$

For the inventory constraints, let $\gamma \in R_i$ index a visit copy of customer i. Since inventory decreases continuously between replenishments, the lower inventory limit is enforced upon arrival, immediately before the current delivery. The upper inventory limit is enforced upon departure, after the delivered quantity has entered the tank. When dispensing occurs over the interval $[a_{i\alpha}, b_{i\alpha}]$, the inventory trajectory can be maintained between these endpoint levels. The resulting constraints are

$$I_i^0 y_{i\alpha} + \sum_{\substack{\gamma \in R_i \\ \gamma < \alpha}} q_{i\gamma} - \rho_i a_{i\alpha} \geq L_i y_{i\alpha} \qquad \forall i \in C,\ \alpha \in R_i, \tag{19}$$

$$I_i^0 y_{i\alpha} + \sum_{\substack{\gamma \in R_i \\ \gamma \leq \alpha}} q_{i\gamma} - \rho_i b_{i\alpha} \leq U_i y_{i\alpha} + Q \sum_{\substack{\gamma \in R_i \\ \gamma < \alpha}} (y_{i\gamma} - y_{i\alpha}) \qquad \forall i \in C,\ \alpha \in R_i, \tag{20}$$

$$I_i^0 + \sum_{\alpha \in R_i} q_{i\alpha} - \rho_i H \geq F_i \qquad \forall i \in C. \tag{21}$$

Constraints (9) and (10) impose a common order on the active visit copies. Constraints (19) and (20) use that same order when accumulating earlier deliveries. The routing, timing, load, and inventory variables therefore describe a single replenishment plan and a single associated inventory trajectory.

The final term in constraint (20) relaxes the upper-limit constraint when copy α is inactive. Constraint (9) places all active copies before inactive copies, ensuring that $y_{i\gamma} - y_{i\alpha} \geq 0$ whenever $\gamma < \alpha$.

Because each visit delivers at most Q, define $k_i = \lceil (F_i + \rho_i H - I_i^0)/Q \rceil$ as a lower bound on the number of visits required by customer i. The instance is infeasible if $k_i > n_i$. Otherwise, the valid fixings $y_{i\alpha} = 1$ for all $\alpha = 1, \ldots, k_i$ fix the corresponding minimum number of active copies.

4. Failure of Pairwise Event Precedence

For each customer $i \in C$, let $\mathcal{A}_i$ denote the set of active delivery events. For two distinct events $e, f \in \mathcal{A}_i$, a pairwise formulation may introduce a binary variable o^i_{ef} that equals one when event e precedes event f. We write $e \prec f$ whenever $o^i_{ef} = 1$.

Pairwise antisymmetry imposes $o^i_{ef} + o^i_{fe} = 1$ for all $i \in C$ and distinct $e, f \in \mathcal{A}_i$. Pairwise antisymmetry selects one direction for each pair of events but does not impose transitivity. It therefore permits cyclic precedence relations among three or more events.

When inventory constraints identify earlier deliveries through these pairwise variables, each event may be evaluated using a different implied history. The individual upper-limit constraints can then be satisfied even though no common ordering of the deliveries satisfies all of them simultaneously.

For the four deliveries in the illustrative instance, let $\mathcal{A} = \{0, 1, 2, 3\}$ denote the event set. The selected precedence relations include $0 \prec 1$, $1 \prec 2$, and $2 \prec 0$. These relations form a directed cycle. Consequently, no event has all other deliveries as predecessors, and each upper-limit constraint can omit at least one five-unit delivery.

For any customer $i \in C$ and any three distinct events $e, f, g \in \mathcal{A}_i$, the transitivity inequalities $o^i_{ef} + o^i_{fg} \leq 1 + o^i_{eg}$ exclude such cycles. Adding these inequalities makes the pairwise-precedence formulation infeasible on the instance considered below.

Illustrative instance. Consider a single customer, indexed by 1. Set $H = 10$, $I_1^0 = 90$, $\rho_1 = 1$, $L_1 = 0$, $U_1 = 100$, $F_1 = 100$, and $n_1 = 4$. Let $m = 4$ vehicles be available, each with capacity $Q = 5$. Service time is zero, and the two depot–customer arcs satisfy $\tau_{01} = \tau_{10} = 5$ and $c_{01} = c_{10} = 1$. Any vehicle completing a depot–customer–depot trip by time $H = 10$ must arrive at the customer at time 5. Inventory immediately before delivery is therefore $I_1^0 - \rho_1(5) = 85$, so the tank can receive at most $U_1 - 85 = 15$ units. The terminal constraint requires $I_1^0 + \sum_{\alpha \in R_1} q_{1\alpha} - \rho_1 H \geq F_1$, and hence requires at least 20 units of replenishment. The event-copy formulation is therefore infeasible. The pairwise-precedence formulation admits four deliveries of five units each at time 5, with total travel cost 8. Their combined effect raises inventory to $85 + 20 = 105 > U_1$.

The four depot–customer–depot trips satisfy the fleet, travel-time, and vehicle-capacity constraints, and the aggregate delivery quantity satisfies the terminal inventory requirement. The event-order representation causes the violation. The pairwise formulation evaluates different inventory constraints against incompatible subsets of earlier deliveries, whereas the event-copy formulation evaluates all visits along one common inventory trajectory.

Supporting materials for the case study, including the data, source document, reference solver, and verification instructions, are available in the corresponding folder of the OR-Bench GitHub repository.

References

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